

Standard Tries

1 Definition

Standard Trie

A **standard trie** stores a set of strings using one character per edge or node transition. A root-to-terminal path spells one stored string.

A standard trie follows these rules:

- every node except the root is labelled with a character or prefix position;
- the root does not represent a normal character;
- children of a node follow a fixed canonical order;
- each stored string ends at a terminal or external node.

Path Rule

If a path from the root reads

$$c_1, c_2, \dots, c_k,$$

then the node reached by that path represents the prefix

$$c_1 c_2 \dots c_k.$$

2 Canonical Ordering

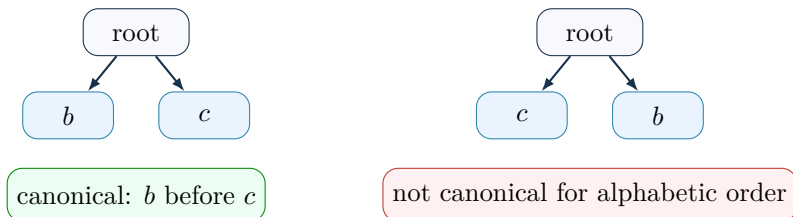
Canonical Ordering

Sibling nodes are stored in a fixed order, usually alphabetic order. This makes the trie representation consistent.

For example,

$$b < c,$$

so the child labelled b should appear before the child labelled c .



3 Example Word Set

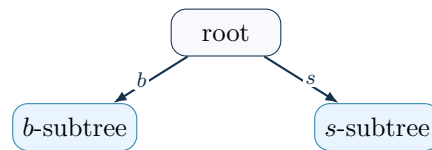
Use the set

$$S = \{\text{bear, bell, bid, bull, buy, sell, stock, stop}\}.$$

The first letter splits the words into two main groups:

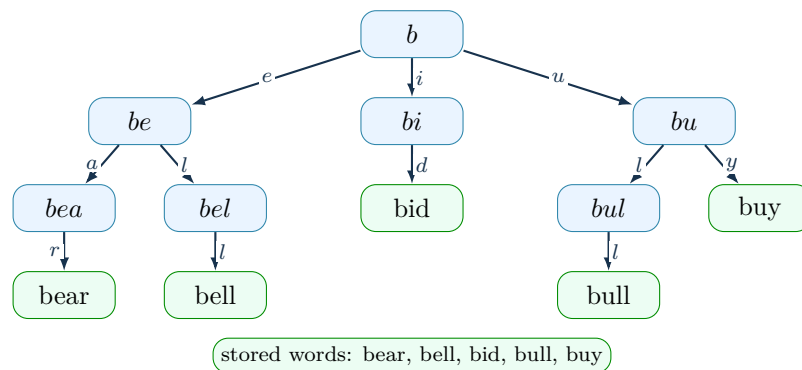
b-words and s-words.

4 Standard-Trie Diagram

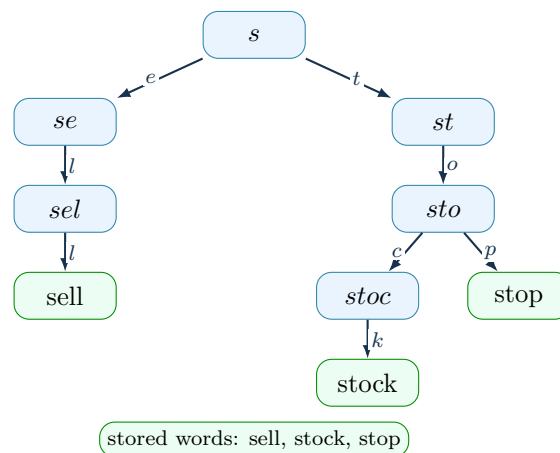


The full trie has two top-level branches. Each branch is expanded below.

b-Subtree



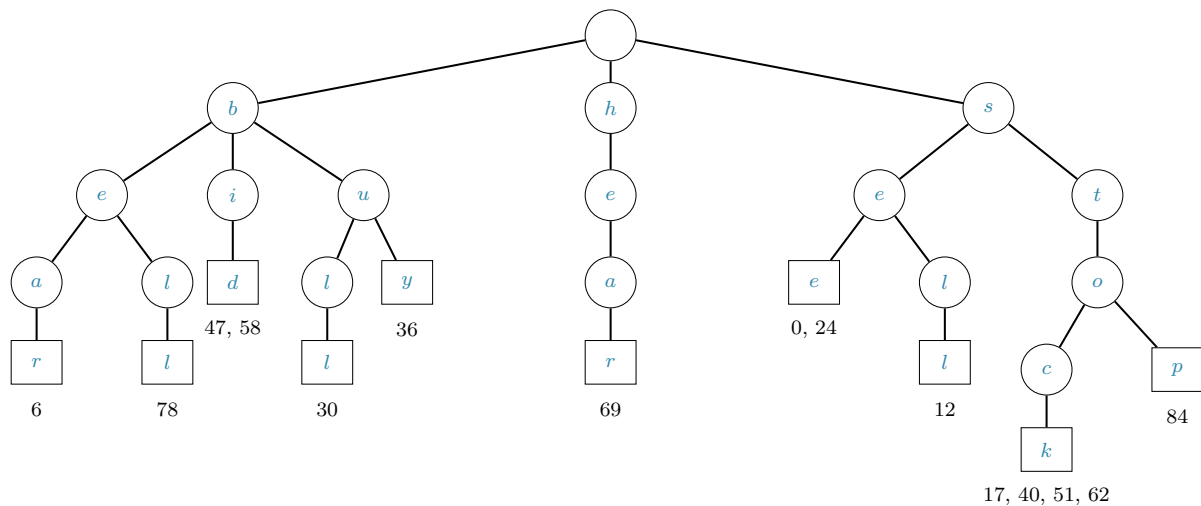
s-Subtree



5 Word-Matching Example

A trie can also store the distinct searchable words from a text. Each leaf stores the starting positions where that word appears.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
s	e	e		a		b	e	a	r	?		s	e	l	l		s	t	o	c	k	!
23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
	s	e	e		a		b	u	l	l	?		b	u	y		s	t	o	c	k	!
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68
	b	i	d		s	t	o	c	k	!		b	i	d		s	t	o	c	k	!	
69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88			
h	e	a	r		t	h	e		b	e	l	l	?		s	t	o	p	!			



Leaf Positions

The number below a leaf is the starting index of that word in the text. For example, stock appears at positions 17, 40, 51, 62.

6 Internal and External Nodes

Internal Node

An **internal node** is a non-root node that is not a leaf. It represents a partial prefix that can still be extended.

Examples from the trie above:

be, bu, sto

are internal prefix nodes.

External Node

An **external node** or leaf represents the end of a stored string.

Examples:

bear, bell, bid, bull, buy, sell, stock, stop.

If a trie stores s strings and each string has its own leaf, then

$$\text{number of external nodes} = s.$$

7 Number of Children

Let

d = size of the alphabet.

Every internal node can have between 1 and d children:

$$1 \leq \text{number of children of an internal node} \leq d.$$

For English lowercase letters,

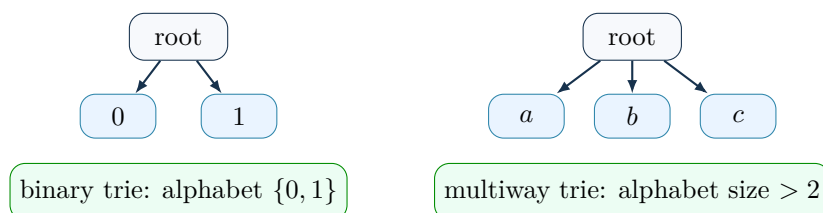
$$d = 26.$$

So a node may have up to 26 child branches.

Important Distinction

The maximum number of children depends on the alphabet size, not on the number of strings. A node in an English-letter trie cannot have more than 26 letter-labelled children.

8 Binary and Multiway Tries



Type	Alphabet	Maximum children per node
Binary trie	$\{0, 1\}$	2
Multiway trie	letters, digits, symbols, or larger alphabets	d , where d is the alphabet size